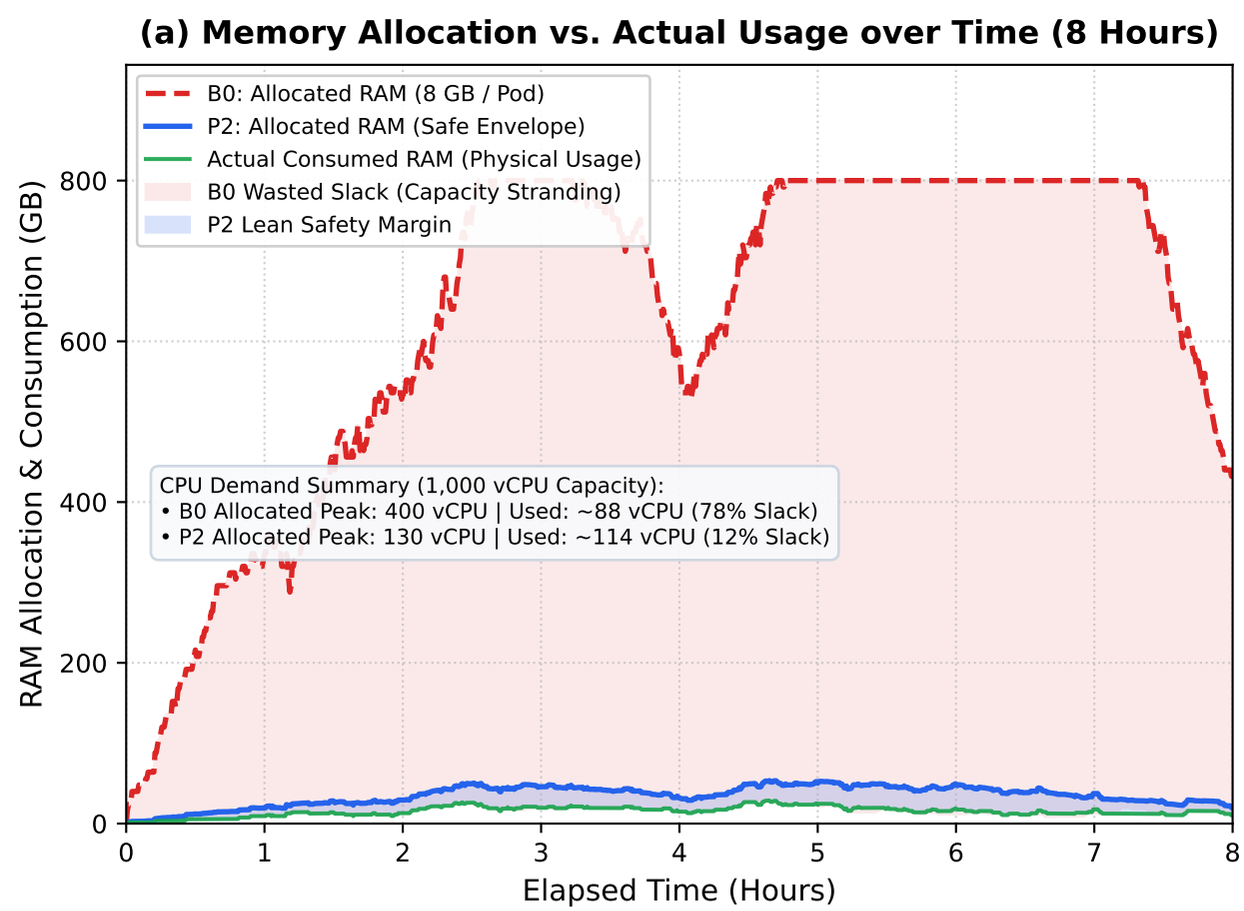
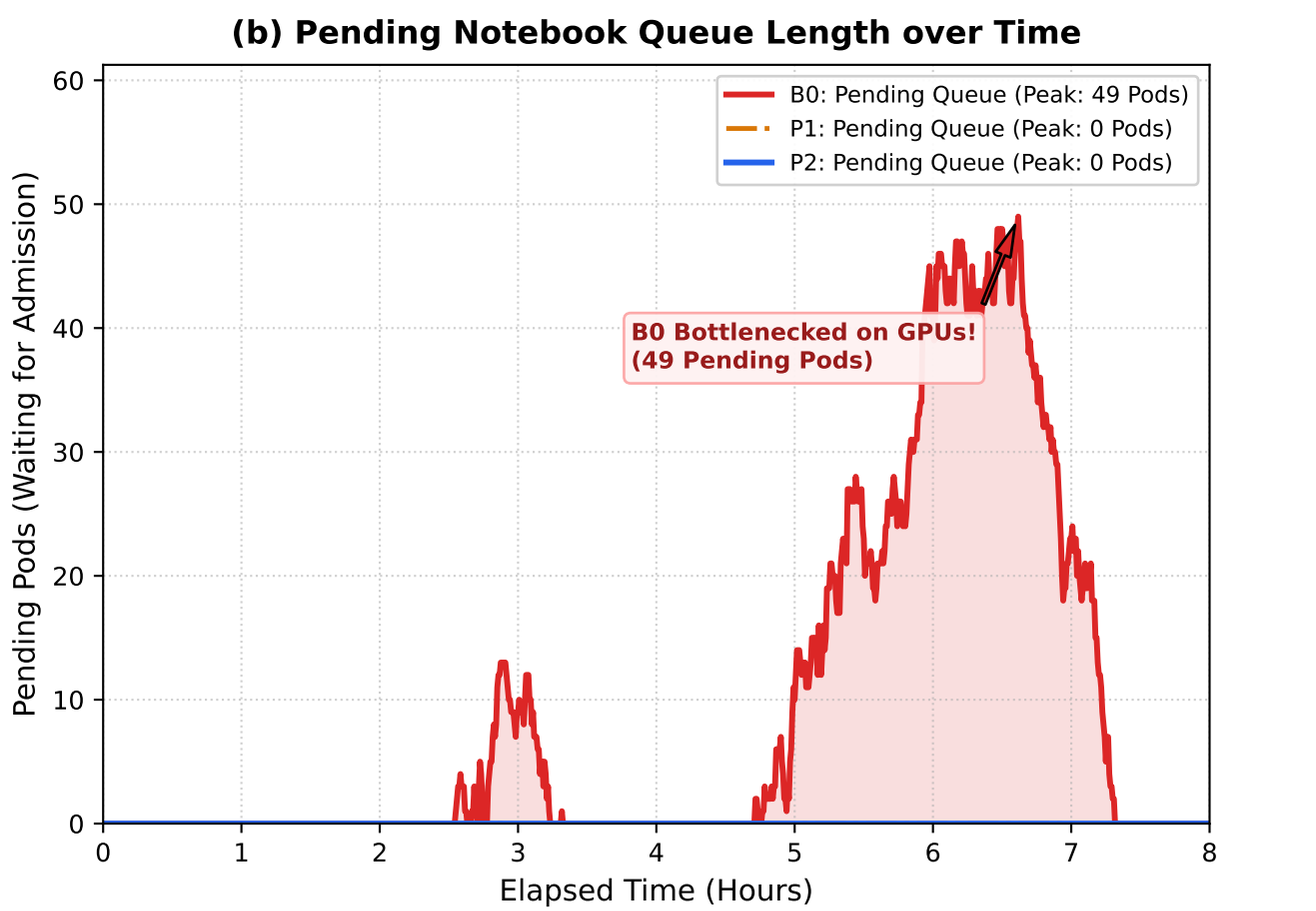
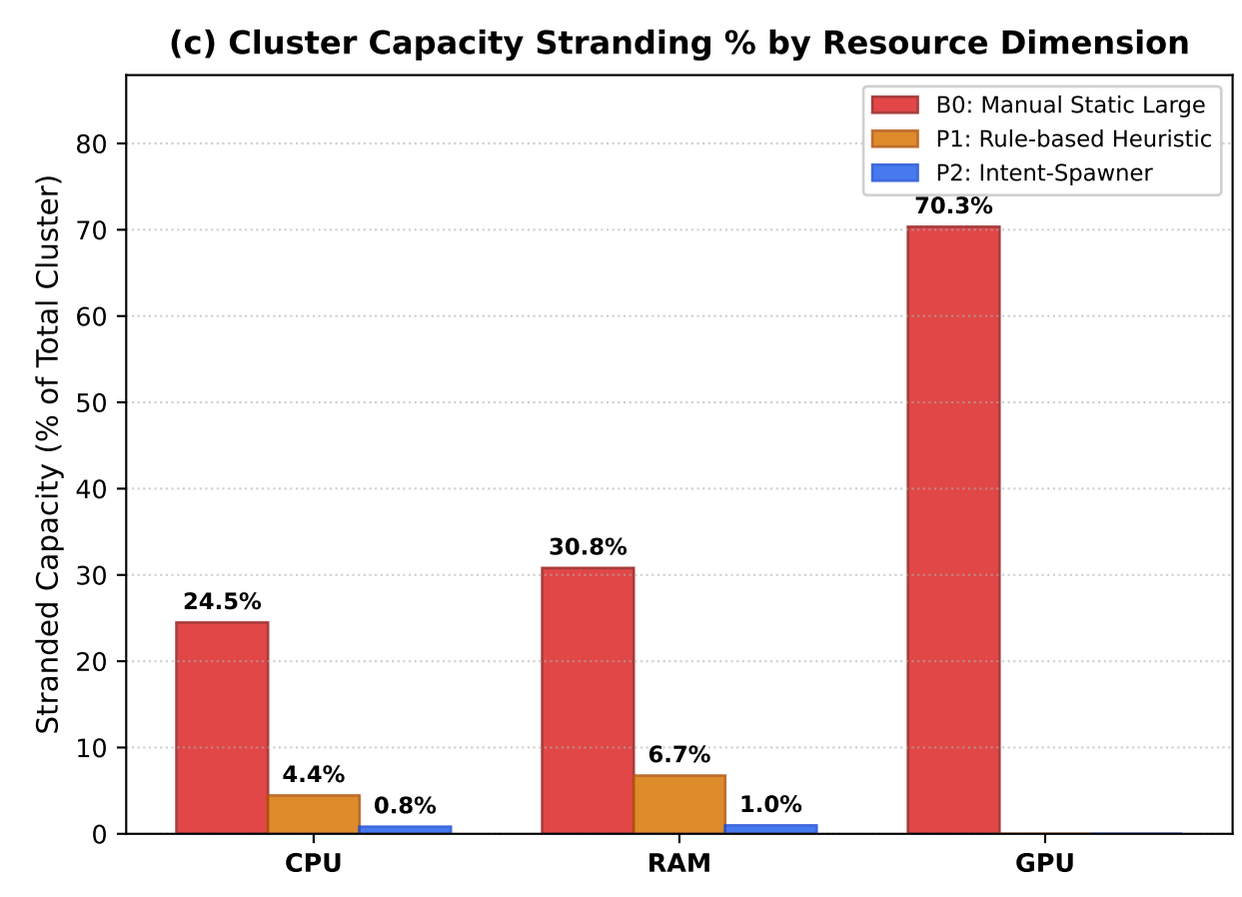
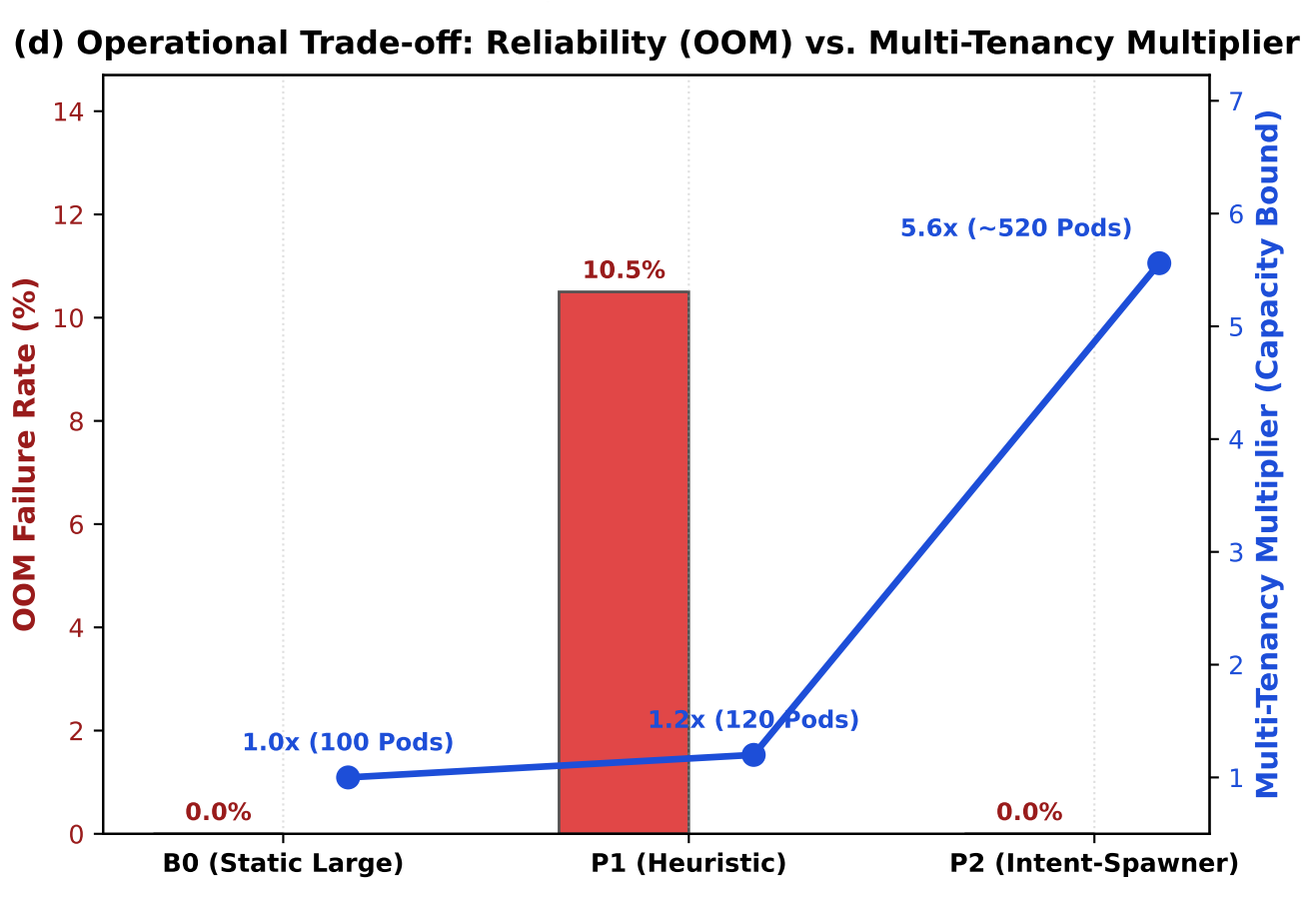

(a) Memory Allocation vs. Actual Usage over Time (8 Hours)
B0: Allocated RAM (8 GB / Pod)
P2: Allocated RAM (Safe Envelope)
Actual Consumed RAM (Physical Usage)
B0 Wasted Slack (Capacity Stranding)
P2 Lean Safety Margin
CPU Demand Summary (1,000 vCPU Capacity):
• B0 Allocated Peak: 400 vCPU | Used: ~88 vCPU (78% Slack)
• P2 Allocated Peak: 130 vCPU | Used: ~114 vCPU (12% Slack)
RAM Allocation & Consumption (GB)
Elapsed Time (Hours)


(b) Pending Notebook Queue Length over Time
B0: Pending Queue (Peak: 49 Pods)
P1: Pending Queue (Peak: 0 Pods)
P2: Pending Queue (Peak: 0 Pods)
B0 Bottlenecked on GPUs!
(49 Pending Pods)
Pending Pods (Waiting for Admission)
Elapsed Time (Hours)


(c) Cluster Capacity Stranding % by Resource Dimension
B0: Manual Static Large
P1: Rule-based Heuristic
P2: Intent-Spawner
24.5%
4.4%
0.8%
30.8%
6.7%
1.0%
70.3%
CPU
RAM
GPU
Stranded Capacity (% of Total Cluster)


(d) Operational Trade-off: Reliability (OOM) vs. Multi-Tenancy Multiplier
OOM Failure Rate (%)
Multi-Tenancy Multiplier (Capacity Bound)
0.0%
10.5%
0.0%
1.0x (100 Pods)
1.2x (120 Pods)
5.6x (~520 Pods)
B0 (Static Large)
P1 (Heuristic)
P2 (Intent-Spawner)